Emin Yusuf AYDIN

PhD Candidate · Bioengineering Program

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Education

Hacettepe University *Ankara, Türkiye*

PhD Bioengineering Division *07.10.2020 - present*

- Advisor: Assoc.Prof. Dr. Ismail UYANIK
- Thesis: Modeling The Role Of Active Sensing Movements During Refuge Tracking Behavior In Weakly Electric Fish
- GPA: 3.97/4.00

Ankara Yıldırım Beyazıt University *Ankara, Türkiye*

MSc Muscoloskeletal System and Regenerative Medicine *20.01.2017 - 07.07.2020*

- Advisor: Prof. Dr. Murat BOZKURT
- Thesis: Comparison of Co-Use Stromal Vascular Fraction and Bone Marrow Concentrate According to Their Single Use the Effects of the Achilles Tendon
- GPA: 3.91/4.00

Başkent University *Ankara, Türkiye*

BS Physiotherapy and Rehabilitation *31.08.2009 - 12.06.2013*

- GPA: 3.23/4.00

Publications

Articles

Aydın, E. Y., Unlutabak, B. & Uyanik, I. (2025). Flow impairs multisensory tracking and increases active sensing in weakly electric fish. Journal of Experimental Biology, jeb-251478. doi:10.1242/jeb.251478

Karagoz, O. K., Kilic, A., **Aydın, E. Y.**, Ankarali, M. M., & Uyanik, I. (2025). Predictive Uncertainty in State-Estimation Drives Active Sensing in Weakly Electric Fish. Bioinspir. Biomim. 20 016018,.doi:10.1088/1748-3190/ad9534.

Aydın, E. Y., Aşık, M., Aydın, H. M., Çay, N., Gümüşkaya, B., Çağlayan, A., Torabi, A., Yüksel, S., Veizi, E., & Bozkurt, M. (2023). The co-use of stromal vascular fraction and bone marrow concentrate for tendon healing. Current stem cell research & therapy, 18(8), 1150-1159. doi: 10.2174/1574888X18666230221141743.

Yüksel, S., Aşık, M. D., Aydin, H. M., Tönük, E., **Aydın, E. Y.**, & Bozkurt, M. (2022). Fabrication of a multi-layered decellularized amniotic membranes as tissue engineering constructs. Tissue and Cell, 74, 101693. doi:10.1016/j.tice.2021.101693.

Seyhan, R. G., Tunay, V. B., Göksülük, D., **Aydın, E. Y.**, & Ergun, N. (2020). Bedensel engelli yüzücülerde çıkış süresinin gövde esnekliği, aerobik endurans ve anaerobik güç ile ilişkisi. Journal of Exercise Therapy and Rehabilitation, 7(2), 186-192.

Book Chapter

Aydın, E. Y., & Uyanik, I. (2023). A Case Study as a Multisensory Integartion Model: Weakly Electric Fish. Editor:Dinçer PR, Use of Zebrafish as a Model Organism. 1. Edition. Ankara: Türkiye Klinikleri., p.104-9.

Presentations

Full Paper

Aydın, E. Y., Yılmaz, O., Öztürk, M., Doğan, H., & Uyanık, İ. (2026, August 23–28). Real-Time Decomposition of Active Sensing and Tracking Motions via a Structure-Aware Filtering Framework. 23rd IFAC World Congress (IFAC WC 2026), Busan, South Korea (accepted).

Topçakar, S., **Aydın, E. Y.**, Demirel, A., & Uyanık, İ. (2024, 15-18 May). Reshaping Active Sensing via Closing a Feedback Loop Around Free Behavior. 32nd Signal Processing and Communications Applications Conference (SIU) (pp. 1-4). IEEE.

Çatalbaş, B., Elikuru, D., **Aydın, E. Y.,** & Uyanik, İ. (2023, 5-8 July). Tracking the Nodal Point of Weakly Electric Fish Using Artificial Neural Networks. 31st Signal Processing and Communications Applications Conference (SIU) (pp. 1-4). IEEE.

Koç, O., Demirel, A., **Aydın, E. Y.,** İbişoğlu, F., Solmaz, Ş. İ., Arı, K., Idman, A., & Uyanık, İ. (2022, 15-18 May). System Identification of the Target Tracking Behavior of Zebrafish During Rheotaxis. 30th Signal Processing and Communications Applications Conference (SIU) (pp. 1-4). IEEE.

Contributed Presentations

Aydın, E. Y., , & Uyanık, I. (2025, 15-19 November). Adaptation of Active Sensing Responses to Sensory Manipulations in Weakly Electric Fish. SFN Meeting 2025 (NEUROSCIENCE 2025).San Diego,CA,USA.

Aydın, E. Y., Karagoz, O. K., Ankarali, M. M., Demirel, A., & Uyanık, I. (2024, 5-9 October). A fish-in-the-loop comparison of various active sensing models. SFN Meeting 2024 (NEUROSCIENCE 2024).Chicago,IL,USA.

Aydın, E. Y., & Uyanık, I. (2024, 25-29 June). A Fish-in-the-loop System to Study the Underlying Mechanisms of Active Sensing, FENS FORUM 2024, Vienna, Austria.

Aydın, E. Y., Yilmaz, O., Topcakar, S., & Uyanık, I. (2024). Closed-loop Manipulation of Active Sensing Movements of Weakly Electric Fish Integrative and Comparative Biology (Vol. 64,Issue Supplement 1, pp. S29-S29)

Ankarali, M.M., Karagoz, O.K., Kilic, A.,**Aydın, E. Y.,**, & Uyanık, I. (2024). Exploring Motion-Based Active Sensing in Relation to Predictive State-Estimation Uncertainty. Integrative and Comparative Biology (Vol. 64,Issue Supplement 1, pp. S19-S20)

Aydın, E. Y., & Uyanık, I. (2023, 11-15 November). Modeling the Effects of Flow Speed on Smooth-Pursuit Tracking in Active Sensing Movements of Weakly Electric Fish. SFN Meeting 2023 (NEUROSCIENCE 2023). Washington D.C.,USA.

Aydin, E., & Uyanik, I. (2023). Flow Speed Affects the Smooth Pursuit Tracking and Active Sensing Movements of Weakly Electric Fish. Integrative and Comparative Biology (Vol. 63, Issue Supplement 1, pp. S26-S26).

Aydın, E. Y., & Uyanık, I. (2022). Investigation of flow speed on the active sensing movements of weakly electric fish. Anatomy: International Journal of Experimental & Clinical Anatomy, 16.

Elikuru, M. D., **Aydın, E. Y.,** Gürler, N., Çatalbaş, B., & Uyanık, I. (2022). Tracking the anal fin and nodal point of weakly electric fish using deep learning. Anatomy: International Journal of Experimental & Clinical Anatomy, 16.

Mardin, C. Ş., İbişoğlu, F., **Aydın, E. Y.,** Öztaş, M., & Uyanık, İ. (2022). The effects of sensory salience on the refuge tracking behavior of weakly electric fish. Anatomy: International Journal of Experimental & Clinical Anatomy, 16.

Research Experience

Hacettepe University - Dept of Electrical and Electronics Engineering *Ankara, Turkey*
Supervisors: Dr. İsmail Uyanik *2023, Present*
- Projects: Systems Description of Multisensory Integration and Dynamic Sensory Weighting (Project No. 123E155)

Hacettepe University - Dept of Electrical and Electronics Engineering *Ankara, Turkey*
Supervisors: Dr. İsmail Uyanik *2021 - 2023*
- Projects: Behavioral Mechanisms of Active Sensing (Project No. 120E198)

Ankara Yıldırım Beyazıt University - Institute Of Health Sciences *Ankara, Turkey*
Supervisors: Dr. Murat Bozkurt, Dr. Mehmet Doğan Aşık *2019 - 2020*
- Projects: The Effects of Mixed Use of Stromal Vascular Fraction and Bone Marrow Concentrate Compared to İndividual Use on Achilles Tendon (Project No. 119S901)

Conferences

- Neuroscience 2025 (SFN 2025), San Diego, CA, USA, 15-19 November 2025
- Neuroscience 2024 (SFN 2024), Chicago, IL, USA, 5-9 October 2024
- FENS Forum 2024, Vienna, Austria, 25-29 June 2024
- Annual Meeting of SICB 2024, Seattle, WA, USA, 2-6 January 2024
- Neuroscience 2023 (SFN 2023), Washington DC, USA, 11-15 November 2023

- Annual Meeting of SICB 2023, Austin, TX, USA, 3-7 January 2023
- FENS Forum 2022, Paris, France, 9-13 July 2022

Professional Development

Courses

SOAR/BISSCITs Workshop 2026 Selected for a competitive travel award to attend the SOAR/BISSCITs Workshop, where interdisciplinary exchange in bio-inspired sensing and control strengthened my research perspective and international collaborations.

METEOR – Methodologies for teamworking in eco-outwards research (EU-funded, 2025) Completed structured training focused on transversal skill development, interdisciplinary collaboration, and enhanced career preparedness for early-stage researchers.

FENS – Chen Institute – NeuroLéman Summer School 2023 The knowledge of motor movements and sensory systems, mainly through studies on living beings, has immensely enriched my thesis research. The valuable global networking offered fresh perspectives for my field of study and postdoc applications. The lab visits at EPFL provided critical firsthand insights into our strengths and areas for improvement.

Animal Use In Experimental Research 2019 I obtained the mandatory 80-hour certification for performing research involving laboratory animals in Turkey through a course at Medipol University.

Workshops

- Erasmus+ Workshop On Biomechanics, Heidelberg University, 1-5 July 2019
- Erasmus+ Project- 'Emerging Nanotechnology Approaches in Muscoloskeletal Tissue Engineering and Regenerative Medicine',Ankara Yıldırım Beyazıt University , 26-30 November 2018
- Erasmus+ Project-'Bench to Bedside and Back',Utrecht University, 16-20 April 2018
- Erasmus+ Project-Novel Approach: 'Translational Postgraduate Education and Training Programme in Musculoskeletal Regenerative Research', Bologna University, 6-10 November 2017

Technical Skills

Programming and Data Analysis

- **Data Processing:** Signal Processing, Time-Series Analysis, Statistical Modeling
- **Machine Learning:** Supervised and Unsupervised Learning, Neural Networks, Dimensionality Reduction

Experimental and Computational Tools

- **Neuroscience Research:** Electrophysiology, Behavioral Analysis, Motion Tracking
- **Simulation & Modeling:** System Identification, Bioinspired Control Systems
- **Hardware & Embedded Systems:** Arduino, Raspberry Pi, STM32, Data Acquisition Systems

Software and Platforms

- **Scientific Computing:** MATLAB,Python
- **Documentation & Typesetting:** LaTeX

Awards, Fellowships, & Grants

2021-Present	**2211A- National PhD Scholarship Program**, The Scientific and Technological Research Council of Türkiye (TÜBİTAK)	*₺ 32,500/ monthly*
2023	**Full Stipend for FENS – Chen Institute – NeuroLéman Summer School 2023**, Federation of European Neuroscience Societies (FENS) Grants	*€ 540*
2026	**Full Stipend for SOAR/BISSCITs Workshop 2026**, SOAR/BISSCITs Workshop 2026 Committee	*€ 1500*

Professional Memberships

- Federation of European Neuroscience Societies
- Neuroscience Society of Turkey